

Mathematical Logic Exercise Series

Exercise 1 (Intermediate). Let P : "It is raining" and Q : "The streets are wet". Translate the following symbolic propositions into English sentences:

1. $P \wedge Q$
2. $P \Rightarrow Q$
3. $\neg Q \Rightarrow \neg P$
4. $P \vee \neg Q$

Exercise 2 (Intermediate). Construct truth tables for the following compound propositions:

1. $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$
2. $\neg(P \wedge Q) \vee (P \Rightarrow Q)$
3. $(P \vee Q) \Rightarrow (P \wedge Q)$

Exercise 3 (Intermediate). Use truth tables to verify the following logical equivalences:

1. $P \Rightarrow Q \equiv \neg P \vee Q$
2. $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$
3. $P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$

Exercise 4 (Intermediate). Let $P(x)$: " x is even" and $Q(x)$: " x is divisible by 3", where $x \in \mathbb{Z}$. Translate the following symbolic statements into English:

1. $\forall x, P(x) \Rightarrow \neg Q(x)$
2. $\exists x, P(x) \wedge Q(x)$
3. $\forall x, \neg(P(x) \wedge Q(x))$
4. $\exists x, \neg P(x) \wedge \neg Q(x)$

Exercise 5 (Intermediate). Write the negation of the following statements in both symbolic form and clear English:

1. "All students love mathematics."
2. "There exists a real number whose square is negative."
3. "For every integer n , if n is even then n^2 is even."
4. "There is a triangle that is both right-angled and equilateral."

Exercise 6 (Intermediate). Use De Morgan's Laws to simplify the following expressions:

1. $\neg(P \wedge \neg Q)$
2. $\neg(\neg P \vee Q)$
3. $\neg(P \Rightarrow Q)$
4. $\neg(P \Leftrightarrow Q)$

Exercise 7 (Intermediate). Determine whether the following pairs of statements are logically equivalent:

1. $P \Rightarrow (Q \wedge R)$ and $(P \Rightarrow Q) \wedge (P \Rightarrow R)$
2. $P \Rightarrow (Q \vee R)$ and $(P \Rightarrow Q) \vee (P \Rightarrow R)$
3. $\neg(P \Rightarrow Q)$ and $P \Rightarrow \neg Q$

Exercise 8 (Intermediate-Hard). Write the contrapositive, converse, and inverse of the following implications:

1. "If a number is divisible by 6, then it is divisible by 3."
2. "If a triangle is equilateral, then all its angles are equal."
3. "If $x^2 > 4$, then $x > 2$ or $x < -2$."

Exercise 9 (Intermediate-Hard). Prove using direct proof: "The sum of any two rational numbers is rational."

Exercise 10 (Intermediate-Hard). Prove using contrapositive: "If n^2 is odd, then n is odd."

Exercise 11 (Intermediate-Hard). Prove using contradiction: "There are infinitely many prime numbers."

Exercise 12 (Intermediate-Hard). Find a counterexample to disprove the statement: "For all real numbers x and y , if $x^2 = y^2$ then $x = y$."

Exercise 13 (Intermediate-Hard). Let the domain be all real numbers. Determine the truth value of:

1. $\forall x \exists y, x + y = 0$
2. $\exists y \forall x, x + y = 0$
3. $\forall x \exists y, x \cdot y = 1$
4. $\exists y \forall x, x \cdot y = x$

Exercise 14 (Hard). Prove by induction: For all positive integers n ,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Exercise 15 (Hard). Prove by induction: For all positive integers n , $n^3 - n$ is divisible by 3.

Exercise 16 (Hard). Prove by induction: For all integers $n \geq 4$, $2^n < n!$

Exercise 17 (Hard). Prove using logical equivalences that the following is a tautology:

$$(P \Rightarrow Q) \vee (Q \Rightarrow P)$$

Exercise 18 (Hard). Let $P(x, y)$: " x loves y ", where the domain is all people. Translate the following English statements into symbolic logic:

1. "Everybody loves somebody."
2. "Somebody is loved by everybody."
3. "There is somebody who loves nobody."
4. "Nobody loves everybody."

Exercise 19 (Advanced). Prove that $\sqrt[3]{2}$ is irrational.

Exercise 20 (Advanced). Prove that there exist irrational numbers a and b such that a^b is rational.

Exercise 21 (Advanced). Use proof by contradiction to show that there are no positive integers x and y such that $x^2 - y^2 = 1$.

Exercise 22 (Advanced). Prove that for any sets A and B , $(A \cap B)^c = A^c \cup B^c$ using logical equivalences.

Exercise 23 (Advanced). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Using quantifiers, write the formal definition that:

1. f is increasing
2. f is bounded above
3. f is continuous at a point c
4. f is uniformly continuous